Poster 1 — Who wants to be rich?

DOK 3

Data

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. For each superpower, what percent of its fans chose Rich? Write a fraction and a percent. Which total is the denominator, and why?

b. Draw a segmented bar chart: one 100% bar per superpower, split into Famous / Happy / Healthy / Rich. Whole percents are fine. Add a key and a percent axis.

c. Verdict. Do Freeze Time fans stand out for wanting to be rich? Compare their percent with the other four groups and with all 433 students.

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 2 — The flipped headline

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Find P(Fly | Famous) and P(Famous | Fly) as fractions and percents. Which total is the denominator each time, and why?

b. Shade the "given" group on two copies of the table, one for each probability. Circle the cell the two share.

c. Verdict on each half. Is "most" right in the first sentence? Does the second sentence follow from the first?

d. Repair. Rewrite the headline as two true sentences, each with its probability.

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 3 — Can't happen together, so unrelated?

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

b. Are they independent? Compare P(Famous) with P(Famous | Telepathy).

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

d. Prove it in general. Two events can each happen, and they are mutually exclusive. Explain why they can never be independent. (What does independence say P(A and B) must equal?)

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive. Write the check under it.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 6 (spare) — Rigged coin?

DOK 3

Data

You make the data: 20 flips of a fair coin on the TI-84, where 1 = heads. randInt(0,1,20) STO→ L1, then read the flips down L1 in STAT → EDIT. (randInt is under MATH → PROB.)

The claim you are judging

"I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

c. Estimate P(longest run is 6 or more) from your dotplot.

d. Verdict. First decide: how rare would 6 in a row have to be before you believed the coin was rigged? Then compare your estimate with that bar. Is 6 in a row convincing evidence?

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

Our verdict in one sentence (show your teacher before it goes on the poster):